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Derivative Rules Cheat Sheet

Complete derivative rules cheat sheet with power rule, product rule, quotient rule, chain rule, and all differentiation formulas. Free PDF download for calculus students.

Basic Rules

NAME	FORMULA	NOTES
Constant Rule	$\frac{d}{dx}(c) = 0$	Derivative of a constant is zero
Power Rule	$\frac{d}{dx}(x^n) = nx^{n-1}$	Bring down the exponent, reduce by 1
Constant Multiple	$\frac{d}{dx}(cf(x)) = c \cdot f'(x)$	Constants factor out
Sum Rule	$\frac{d}{dx}(f(x) + g(x)) = f'(x) + g'(x)$	Derivative of sum is sum of derivatives
Difference Rule	$\frac{d}{dx}(f(x) - g(x)) = f'(x) - g'(x)$	Derivative of difference is difference of derivatives

Product & Quotient Rules

NAME	FORMULA	NOTES
Product Rule	$\frac{d}{dx}(f(x) \cdot g(x)) = f'(x)g(x) + f(x)g'(x)$	First times derivative of second plus second times derivative of first
Quotient Rule	$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}$	Low d-high minus high d-low over low squared

Chain Rule

NAME	FORMULA	NOTES
Chain Rule	$\frac{d}{dx}(f(g(x))) = f'(g(x)) \cdot g'(x)$	Derivative of outer times derivative of inner
General Power	$\frac{d}{dx}(u^n) = nu^{n-1} \cdot \frac{du}{dx}$	Power rule with chain rule

Trigonometric Derivatives

NAME	FORMULA	NOTES
Sine	$\frac{d}{dx}(\sin(x)) = \cos(x)$	
Cosine	$\frac{d}{dx}(\cos(x)) = -\sin(x)$	
Tangent	$\frac{d}{dx}(\tan(x)) = \sec^2(x)$	
Cotangent	$\frac{d}{dx}(\cot(x)) = -\csc^2(x)$	
Secant	$\frac{d}{dx}(\sec(x)) = \sec(x)\tan(x)$	
Cosecant	$\frac{d}{dx}(\csc(x)) = -\csc(x)\cot(x)$	

Inverse Trigonometric Derivatives

NAME	FORMULA	NOTES
Arcsine	$\frac{d}{dx}(\arcsin(x)) = \frac{1}{\sqrt{1-x^2}}$	
Arccosine	$\frac{d}{dx}(\arccos(x)) = -\frac{1}{\sqrt{1-x^2}}$	
Arctangent	$\frac{d}{dx}(\arctan(x)) = \frac{1}{1+x^2}$	

Exponential & Logarithmic

NAME	FORMULA	NOTES
Natural Exponential	$\frac{d}{dx}(e^x) = e^x$	
General Exponential	$\frac{d}{dx}(a^x) = a^x \ln(a)$	
Natural Log	$\frac{d}{dx}(\ln(x)) = \frac{1}{x}$	
General Log	$\frac{d}{dx}(\log_a(x)) = \frac{1}{x \ln(a)}$	